

概率论期末复习

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Problem

(基于乘积测度构造独立随机变量, 有限个的情况)

给定 n 个分布函数列 $\{F_k\}_{k=1}^n$. 证明: 存在概率空间 $(\Omega, \mathcal{F}, P)$ 及其上的 n 个随机变量 $\{\xi_k, 1 \leq k \leq n\}$ 使得 $\{\xi_k\}_{k=1}^n$ 独立且满足 $F_{\xi_k}(x) = F_k(x)$.

Proof

对 F_k , 存在 $(\Omega_k, \mathcal{F}_k, P_k)$ 及 $\mathcal{F}_k$ 上的随机变量 $\bar{\xi}_k$ 使得 $F_{\bar{\xi}_k}(x) = F_k(x), k = 1, \dots, n$. 利用测度扩张定理与乘积测度的知识, 定义

$$\Omega := \prod_{k=1}^n \Omega_k, \mathcal{F} := \prod_{k=1}^n \mathcal{F}_k := \sigma\left(\prod_{k=1}^n A_k, A_k \in \mathcal{F}_k\right), \mathbb{P}(\cdot) = \prod_{k=1}^n P_k$$

对乘积概率空间 $(\Omega, \mathcal{F}, \mathbb{P})$ 以及 $\omega = (\omega_1, \dots, \omega_n) \in \Omega$, 定义随机变量

$$\xi_k(\omega) := \bar{\xi}_k(\omega_k), k = 1, 2, \dots, n$$

于是得到 n 个随机变量 $\{\xi_k\}$, 并且满足 $F_{\xi_k} = F_{\bar{\xi}_k} = F_k$, 下证其互相独立.

事实上, 对任意 $B_k \in \mathcal{B}(\mathbb{R})$ 和 ω , 有

$$\begin{aligned} \mathbb{P}\left(\bigcap_{k=1}^n \{\omega : \xi_k(\omega) \in B_k\}\right) &= \mathbb{P}\left(\prod_{k=1}^n \{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\}\right) \\ &= \prod_{k=1}^n P_k(\{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\}) \end{aligned}$$

其中 $\prod$ 和 $\mathbb{P}$ 能交换的原因是: 每个 $A_k := \{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\}$ 均为 $\mathcal{F}_k$ 中的事件, 从而根据乘积测度的定义即知. 最后, 对于 A_k , 将上式所有 $i \neq k$ 的 B_i 取为 $\mathbb{R}$, 得

$$\text{LHS} = \mathbb{P}(\{\omega : \xi_k(\omega) \in B_k\}) = \text{RHS} = P_k(\{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\})$$

再代回上式即得

$$\mathbb{P}\left(\bigcap_{k=1}^n \{\omega : \xi_k(\omega) \in B_k\}\right) = \prod_{k=1}^n \mathbb{P}(\{\omega : \xi_k(\omega) \in B_k\})$$

这就说明了 $\{\xi_k\}_{k=1}^n$ 为 $(\Omega, \mathcal{F}, \mathbb{P})$ 上一组独立的随机变量. □

Problem

(可数个的情况) 给定可列个分布函数列 $\{F_k\}_{k=1}^\infty$. 证明: 存在概率空间 $(\Omega, \mathcal{F}, P)$ 及其上的可列个随机变量 $\{\xi_k, k \geq 1\}$ 使得 $\{\xi_k\}_{k=1}^\infty$ 独立且满足 $F_{\xi_k}(x) = F_k(x)$.

Proof

定义:

$$\Omega := \prod_{k=1}^{\infty} \Omega_k := \{\omega : (\omega_1, \dots, \omega_k, \dots), \omega_k \in \Omega_k\}, \quad \mathcal{F} := \prod_{k=1}^{\infty} \mathcal{F}_k := \sigma(\mathcal{C})$$

其中 $\mathcal{C}$ 表示柱集类, 即

$$\mathcal{C} := \left\{ \prod_{k \in \mathcal{J}} A_k \times \prod_{j \notin \mathcal{J}} \Omega_j, \forall \mathcal{J} \subset \mathbb{Z}^+ \text{ 为有限集}, A_k \in \mathcal{F}_k, k \in \mathcal{J} \right\}$$

再定义

$$\mathbb{P} \left(\prod_{k \in \mathcal{J}} A_k \times \prod_{j \notin \mathcal{J}} \Omega_j \right) := \prod_{k \in \mathcal{J}} P_k(A_k), \forall \mathcal{J} \subset \mathbb{Z}^+ \text{ 为有限集}, A_k \in \mathcal{F}_k, k \in \mathcal{J}$$

对乘积概率空间 $(\Omega, \mathcal{F}, \mathbb{P})$ 以及 $\omega = (\omega_1, \dots, \omega_k, \dots) \in \Omega$, 定义随机变量

$$\xi_k(\omega) := \bar{\xi}_k(\omega_k), \forall k \geq 1$$

于是得到可列个个随机变量 $\{\xi_k\}_{k=1}^{\infty}$, 并且满足 $F_{\xi_k} = F_{\bar{\xi}_k} = F_k$, 下证其互相独立. 根据定义, 只需对任意的 $\{\xi_k\}_{k \in \mathcal{J}}, \mathcal{J} \subset \mathbb{Z}^+$ 为有限集来证明独立性即可.

事实上, 对任意 $B_k \in \mathcal{B}(\mathbb{R})$ 和 ω , 有

$$\begin{aligned} \mathbb{P} \left(\bigcap_{k \in \mathcal{J}} \{\omega : \xi_k(\omega) \in B_k\} \right) &= \mathbb{P} \left(\prod_{k \in \mathcal{J}} \{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\} \times \prod_{j \notin \mathcal{J}} \Omega_j \right) \\ &= \prod_{k \in \mathcal{J}} P_k(\{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\}) \end{aligned}$$

同理, 对特定的 $A_k := \{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\}$, 将上式所有 $i \neq k$ 的 B_i 取为 $\mathbb{R}$, 得

$$\text{LHS} = \mathbb{P}(\{\omega : \xi_k(\omega) \in B_k\}) = \text{RHS} = P_k(\{\omega_k : \bar{\xi}_k(\omega_k) \in B_k\})$$

再代回上式即得

$$\mathbb{P} \left(\bigcap_{k \in \mathcal{J}} \{\omega : \xi_k(\omega) \in B_k\} \right) = \prod_{k \in \mathcal{J}} \mathbb{P}(\{\omega : \xi_k(\omega) \in B_k\})$$

这就说明了 $\{\xi_k\}_{k=1}^{\infty}$ 为 $(\Omega, \mathcal{F}, \mathbb{P})$ 上一列独立的随机变量. □

Problem

设 ξ 为概率空间 $(\Omega, \mathcal{F}, P)$ 上的可积随机变量, 令 $\{A_k\} \subset \mathcal{F}$ 且 $\sum_{k=1}^{\infty} A_k = \Omega$, 定义

$$\eta(\omega) = \sum_{k=1}^{\infty} E(\xi | A_k) I_{A_k}(\omega), \quad \mathcal{C} = \{\emptyset, A_1, A_2, \dots\}, \quad \mathcal{G} = \sigma(\mathcal{C})$$

证明: (1) η 是 $\mathcal{G}$ 可测的;

(2) $\forall A \in \mathcal{G}$, 有 $\int_A \eta P(d\omega) = \int_A \xi P(d\omega)$;

(3) 满足 (1), (2) 的随机变量在 a.s. 意义下唯一.

Proof

(1) is trivial for η is a composition of simple $\mathcal{G}$ -meas, we only need to proof (2).

Define

$$\tilde{\mathcal{G}} = \left\{ A \in \mathcal{G} : \int_A \eta P(d\omega) = \int_A \xi P(d\omega) \right\}$$

We claim:

- $\mathcal{C} \subset \tilde{\mathcal{G}}$:

It's trivial that $\emptyset \in \tilde{\mathcal{G}}$, for A_k , we have

$$\begin{aligned} \int_{A_k} \eta P(d\omega) &= \sum_{i=1}^{\infty} E(\xi | A_i) \int_{A_k} I_{A_i}(\omega) P(d\omega) \\ &\stackrel{A_i \cap A_j = \emptyset, i \neq j}{=} E(\xi | A_k) P(A_k) \\ &= \int_{A_k} \xi P(d\omega) \implies A_k \in \tilde{\mathcal{G}} \end{aligned}$$

- $\tilde{\mathcal{G}}$ is a λ -system:

- $\Omega \in \tilde{\mathcal{G}}$:

$$\begin{aligned} \int_{\Omega} \eta P(d\omega) &= \sum_{i=1}^{\infty} E(\xi | A_i) \int_{\Omega} I_{A_i}(\omega) P(d\omega) = \sum_{i=1}^{\infty} E(\xi | A_i) P(A_i) \\ &= \sum_{i=1}^{\infty} \int_{A_i} \xi P(d\omega) = \int_{\Omega} \xi P(d\omega) (< \infty) \end{aligned}$$

- $A, B \in \tilde{\mathcal{G}}$ and $A \subset B$, then $B - A \in \tilde{\mathcal{G}}$:

$$\begin{aligned} \int_{B-A} \eta P(d\omega) &= \int_B \eta P(d\omega) - \int_A \eta P(d\omega) \\ &= \int_B \xi P(d\omega) - \int_A \xi P(d\omega) = \int_{B-A} \xi P(d\omega) \end{aligned}$$

- $\{B_k\} \subset \tilde{\mathcal{G}}$ with $\{B_k\} \uparrow$, then $\bigcup_k B_k \in \tilde{\mathcal{G}}$:

$$\int_{\bigcup_k B_k} \eta P(d\omega) = \int_{\Omega} \eta I_{\bigcup_k B_k} P(d\omega) = \int_{\Omega} \lim_{n \rightarrow \infty} \eta I_{B_n} dP(d\omega)$$

$$\stackrel{\text{DCT}}{=} \lim_{n \rightarrow \infty} \int_{\Omega} \eta I_{B_n} P(d\omega) = \lim_{n \rightarrow \infty} \int_{\Omega} \xi I_{B_n} P(d\omega)$$

$$\stackrel{\text{DCT}}{=} \int_{\Omega} \xi I_{\cup_k B_k} P(d\omega) = \int_{\cup_k B_k} \xi P(d\omega)$$

The DCT is used because $|\eta I_{B_n}| \leq |\eta|$ and η is integrable on Ω .

$\mathcal{C}$ is a π -system, thus we have $\mathcal{G} = \sigma(\mathcal{C}) = \lambda(\mathcal{C}) \subset \tilde{\mathcal{G}} \implies \tilde{\mathcal{G}} = \mathcal{G}$.

(3) 设 ξ_1, ξ_2 为满足条件的两个随机变量, 记 $\Lambda = \{\omega : \xi_1(\omega) > \xi_2(\omega)\}$, 则 $\Lambda \in \mathcal{G}$ 且

$$\Lambda = \bigcup_{n=1}^{\infty} \Lambda_n, \quad \Lambda_n := \left\{ \omega : \xi_1(\omega) - \xi_2(\omega) > \frac{1}{n} \right\} \in \mathcal{G}$$

由于 $\int_A (\xi_1 - \xi_2) P(d\omega) = 0, \forall A \in \mathcal{G}$, 取 $A = \Lambda_n$ 即知

$$0 = \int_{\Lambda_n} (\xi_1 - \xi_2) P(d\omega) \geq \frac{1}{n} P(\Lambda_n) \implies P(\Lambda_n) = 0, \forall n \geq 1$$

故 $P(\Lambda) = 0$, 同理 $P(\{\omega : \xi_1(\omega) < \xi_2(\omega)\}) = 0$, 故 $P(\{\omega : \xi_1(\omega) \neq \xi_2(\omega)\}) = 0$, 即 $\xi_1 \stackrel{\text{a.s.}}{=} \xi_2$, 这就验证了唯一性. $\square$

Problem

设 ξ, η 是 $(\Omega, \mathcal{F}, P)$ 上的独立随机变量, $p \geq 1$, 证明:

(1) 若 $E(\xi) = E(\eta) = 0$, 则 $E(|\xi + \eta|^p) \geq \max(E(|\xi|^p), E(|\eta|^p))$;

(2) $E(|\xi + \eta|^p) < +\infty \iff E(|\xi|^p) < +\infty, E(|\eta|^p) < +\infty$.

Proof

(1) 由积分变换定理, 有

$$\begin{aligned} E(|\xi + \eta|^p) &= \iint_{\mathbb{R}^2} |x + y|^p dF_{(\xi, \eta)}(x, y) \stackrel{\text{indep}}{=} \iint_{\mathbb{R}^2} |x + y|^p dF_{\xi}(x) dF_{\eta}(y) \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |x + y|^p dF_{\eta}(y) \right) dF_{\xi}(x) = \int_{\mathbb{R}} E(|x + \eta|^p) dF_{\xi}(x) \\ &\geq \int_{\mathbb{R}} |x|^p dF_{\xi}(x) = E(|\xi|^p) \end{aligned}$$

其中最后一步不等式是因为 $f(x) = |x|^p$ 在 $p \geq 1$ 时为凸函数, 由 Jensen 不等式知

$$E(|x + \eta|^p) = E(f(x + \eta)) \geq f(E(x + \eta)) = f(x + E(\eta)) = f(x) = |x|^p$$

同理可知 $E(|\xi + \eta|^p) \geq E(|\eta|^p)$, 故 $E(|\xi + \eta|^p) \geq \max(E(|\xi|^p), E(|\eta|^p))$.

(2) 充分性由 C_r 不等式立得, 下证必要性. 我们有

$$\begin{aligned}
\infty &> \iint_{\mathbb{R}^2} |x+y|^p dF_{(\xi,\eta)}(x,y) \stackrel{\text{indep}}{=} \iint_{\mathbb{R}^2} |x+y|^p dF_{\xi}(x) dF_{\eta}(y) \\
&= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |x+y|^p dF_{\xi}(x) \right) dF_{\eta}(y) = \int_{\mathbb{R}} E(|\xi+y|^p) dF_{\eta}(y)
\end{aligned}$$

于是必然存在 $y_0 \in \mathbb{R}$ 使得 $E(|\xi+y|^p) < \infty$, 从而由 C_r 不等式知

$$E(|\xi|^p) = E(|\xi+y-y|^p) \leq C_p(E(|\xi+y|^p) + |y|^p) < \infty$$

同理可知 $E(|\eta|^p) < \infty$. □

Problem

设 ξ, η 是概率空间 $(\Omega, \mathcal{F}, P)$ 上的两个随机变量, $\mathcal{G} \subset \mathcal{F}$ 为子 σ 代数, 证明:

- (1) 若 $\xi, \eta \in L^2$, $E(\xi^2|\mathcal{G}) = \eta^2$, $E(\xi|\mathcal{G}) = \eta$, 则 $\xi = \eta$ a.s.
- (2) 若 $\xi, \eta \in L^1$, $E(\xi|\mathcal{G}) = \eta$, $E(\eta|\mathcal{G}) = \xi$, 则 $\xi = \eta$ a.s.

Proof

(1) 我们先证明 $E(\xi E(\eta|\mathcal{G})) = E(\eta E(\xi|\mathcal{G}))$, 这是因为

$$\begin{aligned}
E(\xi E(\eta|\mathcal{G})) &= E(E(\xi E(\eta|\mathcal{G})|\mathcal{G})) \stackrel{E(\eta|\mathcal{G}) \text{ is } \mathcal{G}\text{-meas}}{=} E(E(\xi|\mathcal{G}) E(\eta|\mathcal{G})) \\
E(\eta E(\xi|\mathcal{G})) &= E(E(\eta E(\xi|\mathcal{G})|\mathcal{G})) \stackrel{E(\xi|\mathcal{G}) \text{ is } \mathcal{G}\text{-meas}}{=} E(E(\eta|\mathcal{G}) E(\xi|\mathcal{G})) \\
&\implies E(\xi E(\eta|\mathcal{G})) = E(\eta E(\xi|\mathcal{G}))
\end{aligned}$$

对于本题, 易知 $(\xi - \eta)^2 \in L^1$, 于是只需证 $E((\xi - \eta)^2) = 0$, 事实上, 我们有

$$\begin{aligned}
E(\eta^2) &= E(\eta E(\xi|\mathcal{G})) = E(\xi E(\eta|\mathcal{G})) \stackrel{\eta \text{ is } \mathcal{G}\text{-meas}}{=} E(\xi \eta) \\
&\implies E((\xi - \eta)^2) = 2(E(\eta^2) - E(\xi \eta)) = 0 \iff \xi \stackrel{\text{a.s.}}{=} \eta
\end{aligned}$$

(2) 记 $\mathcal{G}_1 = \sigma(\xi)$, $\mathcal{G}_2 = \sigma(\eta)$ 为子 σ 代数. 考虑一个有界的严格递增的函数 f (例如 $f(\xi) = \arctan \xi$), 由于 $(\xi - \eta)(f(\xi) - f(\eta))$ 有界且恒非负, 故

$$\square := E(\xi - \eta)(f(\xi) - f(\eta)) = 0 \iff \xi \stackrel{\text{a.s.}}{=} \eta$$

事实上, 利用条件数学期望的性质, 我们有

$$\begin{aligned}
\square &= E(\xi f(\xi)) + E(\eta f(\eta)) - E(\xi f(\eta)) - E(\eta f(\xi)) \\
E(\xi f(\eta)) &= E(E(\xi f(\eta)|\mathcal{G}_2)) = E(f(\eta) E(\xi|\mathcal{G}_2)) = E(\eta f(\eta)) \\
E(\eta f(\xi)) &= E(E(\eta f(\xi)|\mathcal{G}_1)) = E(f(\xi) E(\eta|\mathcal{G}_1)) = E(\xi f(\xi))
\end{aligned}$$

于是 $\square = 0$, $\xi \stackrel{\text{a.s.}}{=} \eta$. □

以上为期中考试时重点准备的内容, 暂且予以保留.

0.1 多元正态分布

Problem

设 $\vec{\xi} := (\xi_1, \xi_2, \xi_3)^T \sim N(\vec{a}, B)$, 记 $X = \xi_1 + 2\xi_2 - \xi_3$, $Y = \xi_1 + \xi_2 + \frac{1}{2}\xi_3$, 计算 $E(X|Y)$, $E(Y|X)$, $D(X|Y)$, $D(Y|X)$.

Solution

先回顾二元正态分布的结构: 称以

$$p(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-r^2}} \cdot \exp\left(-\frac{1}{2(1-r^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2r(x-\mu_1)(x-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right)$$

为密度函数的连续型分布为二元正态分布, 其中 $\sigma_1, \sigma_2 > 0, r \in (-1, 1)$.

我们列出一些基本的计算结果. 边缘密度函数:

$$p_1(x) = \int_{-\infty}^{\infty} p(x, y) dy = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}}$$
$$p_2(y) = \int_{-\infty}^{\infty} p(x, y) dx = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(y-\mu_2)^2}{2\sigma_2^2}}$$

条件密度函数:

$$p_{\xi|\eta=y}(x) = \frac{p(x, y)}{p_2(y)}, \quad p_{\eta|\xi=x}(y) = \frac{p(x, y)}{p_1(x)}$$

由此即知已知 $Y = y$ 时 X 的分布为

$$N\left(\mu_1 + r\frac{\sigma_1}{\sigma_2}(y - \mu_2), \sigma_1^2(1-r^2)\right)$$

同理可知已知 $X = x$ 时 Y 的分布, 这是本题需要的关键计算结果. 回到本题, 有

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 1 & \frac{1}{2} \end{pmatrix} \vec{\xi} = A\vec{\xi} \sim N(A\vec{a}, ABA^T) = N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & r\sigma_1\sigma_2 \\ r\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}\right)$$

因此可知

$$E(X|Y) = \mu_1 + r\frac{\sigma_1}{\sigma_2}(Y - \mu_2), \quad D(X|Y) = \sigma_1^2(1-r^2)$$

$$E(Y|X) = \mu_2 + r\frac{\sigma_2}{\sigma_1}(X - \mu_1), \quad D(Y|X) = \sigma_2^2(1-r^2)$$

我能送出去分了吗?

Problem

设 $\xi_1, \dots, \xi_n$ 为 iid, 定义样本均值和样本方差分别为

$$\bar{\xi} := \frac{1}{n} \sum_{k=1}^n \xi_k, \quad S_n^2 := \frac{1}{n-1} \sum_{k=1}^n (\xi_k - \bar{\xi})^2$$

若 $\xi_1 \sim N(a, \sigma^2)$, 求证:

- (1) $\bar{\xi}$ 与 S_n^2 相互独立;
- (2) $\bar{\xi}$ 服从正态分布 $N(a, \frac{\sigma^2}{n})$;
- (3) $\frac{(n-1)S_n^2}{\sigma^2}$ 服从 $n-1$ 个自由度的卡方分布 $\chi^2(n-1)$.

Proof

(1) 易知 $\vec{\xi} = (\xi_1, \dots, \xi_n)^T \sim N((a, \dots, a)^T, \sigma^2 I)$. 考虑 n 阶正交阵

$$C = \begin{pmatrix} \frac{1}{\sqrt{1 \cdot 2}} & \frac{-1}{\sqrt{1 \cdot 2}} & 0 & 0 & \dots & 0 & 0 \\ \frac{1}{\sqrt{2 \cdot 3}} & \frac{1}{\sqrt{2 \cdot 3}} & \frac{-2}{\sqrt{2 \cdot 3}} & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \frac{1}{\sqrt{(n-1)n}} & \dots & \frac{1}{\sqrt{(n-1)n}} & \frac{-(n-1)}{\sqrt{(n-1)n}} \\ \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \dots & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} \end{pmatrix}$$

记 $\vec{\eta} = C\vec{\xi}$, 则 $\vec{\eta}$ 服从正态分布, 协方差阵为 $C\sigma^2 I C^T = \sigma^2 I$, 因此 $\vec{\eta}$ 各分量独立且方差为 σ^2 , 且

$$E(\eta_1) = \dots = E(\eta_{n-1}) = 0, E(\eta_n) = \sqrt{n}a$$

由正交变换的保范性可知

$$\sum_{k=1}^n \eta_k^2 = \|\vec{\eta}\|^2 = \|\vec{\xi}\|^2 = \sum_{k=1}^n \xi_k^2$$

注意到 $\eta_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n \xi_k = \sqrt{n}\bar{\xi}$, 而

$$nS_n^2 = \sum_{j=1}^n (\xi_j - \bar{\xi})^2 = \sum_{k=1}^n \xi_k^2 - n\bar{\xi}^2 = \sum_{k=1}^n \eta_k^2 - \eta_n^2 = \sum_{k=1}^{n-1} \eta_k^2$$

由 $\eta_1, \dots, \eta_n$ 相互独立即知 $\bar{\xi} = \frac{\eta_n}{\sqrt{n}}$ 与 $S_n^2 = \frac{1}{n} \sum_{k=1}^{n-1} \eta_k^2$ 相互独立.

(2) 由 $\bar{\xi} = (\frac{1}{n}, \dots, \frac{1}{n}) \cdot \vec{\xi}$ 即知 $\bar{\xi} \sim N(a, \frac{\sigma^2}{n})$.

(3) 我们有 $\frac{nS_n^2}{\sigma^2} = \sum_{k=1}^{n-1} \left(\frac{\eta_k}{\sigma}\right)^2$, 其中 $\frac{\eta_k}{\sigma} \sim N(0, 1)$ 且互相独立, 由之前作业的计算可知 $\left(\frac{\eta_k}{\sigma}\right)^2$ 服从 $\Gamma(\frac{1}{2}, \frac{1}{2})$ 且为 iid, 故其和 $\frac{nS_n^2}{\sigma^2}$ 服从 $\Gamma\left(\frac{1}{2}, \frac{n-1}{2}\right) = \chi^2(n-1)$. $\square$

Problem

二维随机向量的两个分量均服从正态分布, 该随机向量是否服从二元正态分布?

Solution

不一定. 以下例子出自中文教材 P158 例 3.5.1.

考虑独立同服从 $N(0, 1)$ 的随机变量 (ξ, η) , 定义

$$\zeta = \begin{cases} |\eta|, & \xi > 0 \\ -|\eta|, & \xi < 0 \end{cases}$$

则 $\zeta \sim N(0, 1)$, 但 $(\zeta, \eta), (\zeta, \xi)$ 均不服从二元正态分布.

证明: 当 $x > 0$ 时, ζ 的分布函数为

$$\begin{aligned} P(\zeta < x) &= \frac{1}{2}P(|\eta| < x) + \frac{1}{2}P(-|\eta| < x) \\ &= \frac{1}{2} \int_{-x}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt + \frac{1}{2} = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \Phi(x) \end{aligned}$$

类似可知 $x < 0$ 时 ζ 的分布函数也是 $\Phi(x)$, 这就说明了 $\zeta \sim N(0, 1)$.

(ζ, η) 不服从二元正态分布: 利用全概率公式可知

$$\begin{aligned} P(\zeta + \eta = 0) &= \frac{1}{2}P(\eta + |\eta| = 0) + \frac{1}{2}P(\eta - |\eta| = 0) \\ &= \frac{1}{2}P(\eta < 0) + \frac{1}{2}P(\eta > 0) = \frac{1}{2} \end{aligned}$$

若 (ζ, η) 服从二元正态分布, 则 $\zeta + \eta$ 应服从一维正态分布, 上述概率应为 0. 从而可知 (ζ, η) 不服从二元正态分布.

(ζ, ξ) 不服从二元正态分布: 由 $P(\zeta < 0, \xi > 0) = 0$ 立得.

0.2 四种收敛之间的关系以及反例

Problem

(四种收敛之间的关系)

- $\xi_n \xrightarrow{\text{a.s.}} \xi \implies \xi_n \xrightarrow{\mathbb{P}} \xi \implies \xi_n \xrightarrow{d} \xi;$
- $\xi_n \xrightarrow{L^p} \xi \implies \xi_n \xrightarrow{\mathbb{P}} \xi \implies \xi_n \xrightarrow{d} \xi.$

Proof

- 几乎处处收敛 $\implies$ 依概率收敛:

记不收敛点集合为 E , 我们熟知

$$\xi_n \xrightarrow{\text{a.s.}} \xi \iff P(E) \iff \lim_{k \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} \left\{\omega : |\xi_n - \xi| > \frac{1}{m}\right\}\right) = 0, \forall m$$

从而可知

$$\left\{|\xi_k - \xi| > \frac{1}{m}\right\} \subset \bigcup_{n=k}^{\infty} \left\{|\xi_n - \xi| > \frac{1}{m}\right\} \implies \lim_k P\left\{|\xi_k - \xi| > \frac{1}{m}\right\} = 0, \forall m$$

即 $\xi_n \xrightarrow{\mathbb{P}} \xi$.

- L^p 收敛 $\implies$ 依概率收敛: 由 Chebyshev 不等式可知

$$P\{|\xi_n - \xi| > \varepsilon\} \leq \frac{E(|\xi_n - \xi|^p)}{\varepsilon^p}$$

令 $n \rightarrow \infty$ 即可.

- 依概率收敛 $\implies$ 依分布收敛:

I 如果我们有: 对任意 $x_1 < x < x_2$, $x_1, x, x_2 \in C_{F_\xi}$, 如下 (*) 成立

$$F_\xi(x_1) \leq \varliminf_{n \rightarrow \infty} F_{\xi_n}(x) \leq \overline{\varliminf}_{n \rightarrow \infty} F_{\xi_n}(x) \leq F_\xi(x_2) \quad (*)$$

于是, 令 $x_1 \uparrow x$ 且 $x_2 \downarrow x$ 即得 $\lim_n F_{\xi_n}(x) = F_\xi(x)$.

由定义, 有

$$\begin{aligned} F_\xi(x_1) &= P\{\xi < x_1\} = P\{\xi < x_1, \xi_n < x\} + P\{\xi < x_1, \xi_n \geq x\} \\ &\leq F_{\xi_n}(x) + P\{|\xi_n - \xi| \geq x - x_1\} \implies F_\xi(x_1) \leq \varliminf_n F_{\xi_n}(x) \end{aligned}$$

同理有

$$\begin{aligned} F_{\xi_n}(x) &= P\{\xi_n < x\} = P\{\xi_n < x, \xi < x_2\} + P\{\xi_n < x, x \geq x_2\} \\ &\leq F_\xi(x_2) + P\{|\xi_n - \xi| \geq x_2 - x\} \implies \overline{\varliminf}_n F_{\xi_n}(x) \leq F_\xi(x_2) \end{aligned}$$

于是 (*) 成立. □

Problem

四种收敛的反例?

Example

令 $\Omega = [0, 1)$, $\mathcal{F} = \mathcal{B}(\Omega)$, $dP = dx$.

- $\xi_n \xrightarrow{\mathbb{P}} \xi \not\Rightarrow \xi_n \xrightarrow{\text{a.s.}} \xi$:

$$\eta_{ki} = I_{[\frac{i-1}{k}, \frac{i}{k})}, k \geq 1, i = 1, 2, \dots, k$$

令 $\xi_1 = \eta_{11}, \xi_2 = \eta_{21}, \xi_3 = \eta_{22}, \xi_4 = \eta_{31}, \dots$, 则

$$P(\{\omega : |\xi_n| > \varepsilon\}) = P\left(\left[\frac{i-1}{k_n}, \frac{i}{k_n}\right)\right) = \frac{1}{k_n} \rightarrow 0 (n \rightarrow \infty)$$

i.e. $\xi_n \xrightarrow{\mathbb{P}} \xi$, 但是由定义易见 $\xi_n \not\xrightarrow{\text{a.s.}} \xi$.

注意: 虽然课上 (疑似) 没说, 但是这个例子也说明了 $\xi_n \xrightarrow{L^p} \xi \not\Rightarrow \xi_n \xrightarrow{\text{a.s.}} \xi$.

- $\xi_n \xrightarrow{\text{a.s.}} \xi \not\Rightarrow \xi_n \xrightarrow{L^p} \xi$:

定义

$$\xi_n = \begin{cases} 2^n, & \omega \in [0, \frac{1}{n}) \\ 0, & \omega \in [\frac{1}{n}, 1) \end{cases}, \quad \xi \equiv 0$$

于是 $\xi_n \xrightarrow{\text{a.s.}} \xi$, 但是 $\xi_n \not\xrightarrow{L^p} \xi$, 这是因为

$$E(|\xi_n|^p) = \frac{2^{np}}{n} \rightarrow \infty$$

- $\xi_n \xrightarrow{d} \xi \not\Rightarrow \xi_n \xrightarrow{\mathbb{P}} \xi$:

令 $\{\xi_n, \xi\}$ 为 iid, 且 $\xi \sim \begin{pmatrix} 1 & -1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$.

于是 $F_{\xi_n} = F_{\xi} \Rightarrow \xi_n \xrightarrow{d} \xi$, 但

$$P\{|\xi_n - \xi| > \varepsilon\} = P\{\xi_n = 1, \xi = -1\} + P\{\xi_n = -1, \xi = 1\} = \frac{1}{2} \not\rightarrow 0$$

i.e. $\xi_n \not\xrightarrow{\mathbb{P}} \xi$.

0.3 依分布收敛(弱收敛)

Lemma 0.3.1 (Helly 第一定理)

对任意分布函数列 $\{F_n\}_{n \geq 1}$, 存在子列 $\{F_{n_k}\}$ 以及一个左连续非降函数 F s.t. $F_{n_k} \xrightarrow{\omega} F$.

Lemma 0.3.2 (Skovohod Representation Thm)

设 $\{F_n\}, F$ 为分布函数, 满足 $F_n \xrightarrow{\omega} F$. 则存在概率空间 $(\Omega, \mathcal{F}, P)$ 以及其上的随机变量 $\{\xi_n\}, \xi$ 使得 $F_{\xi_n} = F_n, F_\xi = F$, 并且对任意 $\omega \in \Omega, \xi_n(\omega) \rightarrow \xi(\omega)$.

Remark

证明太长了, 如果考 Helly 第二定理, 我会不加证明地使用这一引理.

Theorem 0.3.3 (Helly 第二定理)

设 $\xi_n \xrightarrow{d} \xi$. 则 $\lim_n \int_{\mathbb{R}} f(x) dF_{\xi_n}(x) = \int_{\mathbb{R}} f(x) dF_\xi(x), \forall f \in C_B := \{f \mid f \text{ 有界连续}\}.$

Proof

由条件, $F_n \xrightarrow{\omega} F$, 故由引理可知存在概率空间 $(\Omega', \mathcal{F}', P')$ 以及其上的随机变量 $\{\eta_n\}, \eta$ 使得 $\eta_n \rightarrow \eta$ 且 $F_{\eta_n} = F_{\xi_n}, F_\eta = F_\xi$. 由于 η_n 点点收敛, 并且 f 连续, 故 $f(\eta_n) \rightarrow f(\eta)$, 由有界性即 DCT 可知 $E(f(\eta_n)) \rightarrow E(f(\eta))$, 即

$$\lim_n \int_{\mathbb{R}} f(x) dF_{\xi_n}(x) = \lim_n \int_{\mathbb{R}} f(x) dF_{\eta_n}(x) = \int_{\mathbb{R}} f(x) dF_\eta(x) = \int_{\mathbb{R}} f(x) dF_\xi(x) \quad \square$$

Theorem 0.3.4 (弱收敛等价于特征函数逐点收敛)

$\xi_n \xrightarrow{d} \xi \iff f_{\xi_n}(t) := E(e^{it\xi_n}) \rightarrow f_\xi(t) := E(e^{it\xi}), \forall t \in \mathbb{R}.$

Proof

$\implies$: 取 $f_t(x) = e^{itx}$ 为有界连续函数, 由 Helly 第二定理知 $E(f_t(\xi_n)) \rightarrow E(f_t(\xi))$.

$\impliedby$: 由 Helly 第一定理, 存在 $F_{n_k} := F_{\xi_{n_k}}, \text{ s.t. } F_{n_k} \xrightarrow{\omega} \hat{F}$, 满足 $\hat{F}$ 非降, 左连续, 取值在 $[0, 1]$ 间.

- Claim 1: $\hat{F}$ 是分布函数.

否则, 我们有

$$(0, 1) \ni a = \hat{F}(\infty) - \hat{F}(-\infty) = \lim_{b \rightarrow \infty, b \in C_{\hat{F}}} \lim_{k \rightarrow \infty} [F_{n_k}(b) - F_{n_k}(-b)]$$

一方面, 由于 $f(0) = 1$ 且 f 一致连续, 对任意 $\varepsilon \in (0, 1 - a)$, 存在 $\delta > 0$ s.t. $|f(t) - 1| < \frac{\varepsilon}{2}, \forall t \in [-\delta, \delta]$. 于是

$$\frac{1}{2\delta} \left| \int_{-\delta}^{\delta} [f(t) - f(0)] dt \right| < \frac{\varepsilon}{2}$$

$$\Rightarrow \frac{1}{2\delta} \left| \int_{-\delta}^{\delta} f(t) dt \right| \geq \frac{1}{2\delta} \int_{-\delta}^{\delta} f(0) dt - \frac{\varepsilon}{2} = 1 - \frac{\varepsilon}{2} > a + \frac{\varepsilon}{2} \quad (*)$$

另一方面, 对于 $b \in C_{\hat{F}}$

$$\begin{aligned} \frac{1}{2\delta} \left| \int_{-\delta}^{\delta} f_{n_k}(t) dt \right| &= \frac{1}{2\delta} \left| \int_{-\delta}^{\delta} \int_{\mathbb{R}} e^{itx} dF_{n_k}(x) dt \right| = \frac{1}{2\delta} \left| \int_{\mathbb{R}} \int_{-\delta}^{\delta} e^{itx} dt dF_{n_k}(x) \right| \\ &\leq \frac{1}{2\delta} \int_{|x|<b} \left| \int_{-\delta}^{\delta} e^{itx} dt \right| dF_{n_k}(x) + \frac{1}{2\delta} \int_{|x|\geq b} \left| \int_{-\delta}^{\delta} e^{itx} dt \right| dF_{n_k}(x) \\ &\leq \int_{|x|<b} dF_{n_k}(x) + \frac{1}{2\delta} \int_{|x|\geq b} \left| \frac{2 \sin tx}{x} \right| dF_{n_k}(x) \\ &\leq \int_{|x|<b} dF_{n_k}(x) + \frac{1}{2\delta} \frac{2}{b} = F_{n_k}(b) - F_{n_k}(-b) + \frac{1}{b\delta} \end{aligned}$$

令 $k \rightarrow \infty$, 我们得到

$$\lim_k \frac{1}{2\delta} \left| \int_{-\delta}^{\delta} f_{n_k}(t) dt \right| \leq \hat{F}(b) - \hat{F}(-b) + \frac{1}{b\delta}$$

由于 $\hat{F}(\infty) = \lim_{b \rightarrow \infty} \hat{F}(b)$, $\hat{F}(-\infty) = \lim_{b \rightarrow \infty} (-b)$, $a = \lim_{b \rightarrow \infty} (\hat{F}(b) - \hat{F}(-b))$. 选取充分大的 $b \in C_{\hat{F}}$ s.t. $\frac{1}{\delta b} < \frac{\varepsilon}{4}$ 并且 $\hat{F}(b) - \hat{F}(-b) < a + \frac{\varepsilon}{4}$. 则

$$\lim_k \frac{1}{2\delta} \left| \int_{-\delta}^{\delta} f_{n_k}(t) dt \right| < a + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = a + \frac{\varepsilon}{2}$$

由于 $f_{n_k}(t) \rightarrow f(t)$ 并且 $|f_{n_k}(t)| \leq 1$, 使用 DCT 可得

$$\frac{1}{2\delta} \left| \int_{-\delta}^{\delta} f(t) dt \right| \leq a + \frac{\varepsilon}{2} \quad (**)$$

于是 (*) 和 (**) 得到矛盾, 从而 $\hat{F}$ 是分布函数.

- Claim 2: $\hat{F} = F$.

我们已经得到 $F_{n_k} \xrightarrow{\omega} \hat{F}$, 于是由 Helly 第二定理可知

$$\int_{\mathbb{R}} e^{itx} dF_{n_k}(x) \rightarrow \int_{\mathbb{R}} e^{itx} d\hat{F}(x)$$

从而由极限的唯一性

$$\int_{\mathbb{R}} e^{itx} dF_{n_k}(x) = f_{n_k}(t) \rightarrow f(t) = \int_{\mathbb{R}} e^{itx} dF \Rightarrow \int_{\mathbb{R}} e^{itx} d\hat{F}(x) = \int_{\mathbb{R}} e^{itx} dF(x)$$

最后由特征函数与分布函数一一对应可知 $\hat{F} = F$.

- Claim 3: $F_n \xrightarrow{\omega} F$.

否则, 存在 $x_0 \in C_F$ s.t. $F_n(x_0) \not\rightarrow F(x_0)$. 由有界性, 取子列 $\{F_{m_k}(x_0)\}$ s.t. $F_{m_k}(x_0) \rightarrow F^*(x_0) \neq F(x_0)$. 对分布函数列 $\{F_{m_k}\}$, 重复以上证明过程可知, 其存在子列 $\{F_{m_{k_j}}\}$ 使得 $F_{m_{k_j}} \xrightarrow{\omega} F$, 但 $F_{m_{k_j}}(x_0) \rightarrow F^*(x_0) \neq F(x_0)$, 矛盾!

综上, $F_n \xrightarrow{\omega} F$. □

Remark

会背下来的.

Problem

经典结论: $X_n \xrightarrow{\mathbb{P}} a \iff X_n \xrightarrow{d} a$, 其中 a 为常数.

Proof

必要性已证, 只需证充分性, 对任意 $\varepsilon > 0$, 我们有

$$\begin{aligned} P(\{|X_n - a| > \varepsilon\}) &= P(\{X_n > a + \varepsilon\}) + P(\{X_n < a - \varepsilon\}) \\ &= 1 + P(\{X_n < a - \varepsilon\}) - P(\{X_n \leq a + \varepsilon\}) \\ &\leq 1 + P(\{X_n < a - \varepsilon\}) - P(\{X_n < a + \varepsilon\}) \end{aligned}$$

令 $n \rightarrow \infty$, 由依分布收敛的定义即知右式趋于 $1 + F_a(a - \varepsilon) - F_a(a + \varepsilon) = 0$. 于是有 $\lim_n P(\{|X_n - a| > \varepsilon\}) = 0, \forall \varepsilon > 0$, 即 $X_n \xrightarrow{\mathbb{P}} a$. $\square$

Problem

设 $\xi_n \xrightarrow{d} \xi, \eta_n \xrightarrow{d} a \in \mathbb{R}$. 证明: $\xi_n + \eta_n \xrightarrow{d} \xi + a$.

Proof

由上一题的结论可知 $\eta_n \xrightarrow{\mathbb{P}} a$. 于是有如下估计

$$P(\xi_n + \eta_n < x) \leq P(\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon) + P(|\eta_n - a| > \varepsilon)$$

对事件 $\{\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon\}$, 有

$$\{\xi_n + \eta_n < x, |\eta_n - a| < \varepsilon\} \subset \{\xi_n < x - a + \varepsilon\}$$

于是取上极限可得

$$\overline{\lim}_n P(\xi_n + \eta_n < x) \leq \lim_n P(\xi_n < x - a + \varepsilon) = P(\xi < x - a + \varepsilon)$$

对于下界, 我们有关系

$$\{\xi_n \leq x - a - \varepsilon\} \subset \{|\eta_n - a| > \varepsilon\} \cup \{\xi_n + \eta_n < x\}$$

因此

$$P(\xi_n + \eta_n < x) \geq P(\xi_n < x - a - \varepsilon) - P(|\eta_n - a| > \varepsilon)$$

取下极限可得

$$\underline{\lim}_n P(\xi_n + \eta_n < x) \geq \lim_n P(\xi_n < x - a - \varepsilon) = P(\xi < x - a - \varepsilon)$$

取 $x \in C_{\xi+a}$, 于是 $x - a \in C_\xi$, 令上述两个式子中 $\varepsilon \rightarrow 0$, 即得

$$\lim_n P(\xi_n + \eta_n < x) = P(\xi + a < x)$$

也即 $\xi_n + \eta_n \xrightarrow{d} \xi + a$. $\square$

Remark

同样的证明方法给出: $\xi_n \xrightarrow{d} \xi, \eta_n \xrightarrow{d} a \neq 0$, 则 $\xi_n \eta_n \xrightarrow{d} a\xi$. $a = 0$ 的情形见下.

Problem

设 $\xi_n \xrightarrow{d} \xi$, $\eta_n \xrightarrow{d} 0$. 证明: $\xi_n \eta_n \xrightarrow{d} 0$.

Proof

首先有 $\eta_n \xrightarrow{\mathbb{P}} 0$, 要证 $\xi_n \eta_n \xrightarrow{d} 0$, 只需证 $\xi_n \eta_n \xrightarrow{\mathbb{P}} 0$. 每个 ξ_n 以及 ξ 的不连续点是可列的, 因此这些不连续点集的并也是可列的. 因此对固定的 $\varepsilon > 0$, 可取充分大的 $A, -A \in C_{F_\xi} \cup \left(\bigcup_{n=1}^{\infty} C_{F_{\xi_n}}\right)$, 且 $P(\{|\xi| > A\}) < \varepsilon$. 于是有如下估计

$$\begin{aligned}
 P(\{|\xi_n \eta_n| > \delta\}) &\leq P(\{|\xi_n| > A\}) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\
 &= 1 - F_{\xi_n}(A+0) + F_{\xi_n}(-A) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\
 &= 1 - F_{\xi_n}(A) + F_{\xi_n}(-A) + P\left(\left\{|\eta_n| \geq \frac{\delta}{A}\right\}\right) \\
 &\xrightarrow{n \rightarrow \infty} 1 - F_\xi(A) + F_\xi(-A) = P(\{|\xi| > A\}) < \varepsilon
 \end{aligned}$$

令 $\varepsilon \rightarrow 0$ 即得 $\lim_n P(\{|\xi_n \eta_n| > \delta\}) = 0$, 综上命题得证. □

0.4 大数定律

先看简单的弱大数定律

Theorem 0.4.1 若 $\{\xi_n\}$ 满足 $\frac{1}{n^2}D(S_n) \rightarrow 0$ (称为 Markov 条件), 则 WLLN 成立.

Proof

由 Chebyshev 不等式 (易证) 可知

$$P\left(\left\{\omega: \left|\frac{1}{n}(S_n - E(S_n))\right| > \varepsilon\right\}\right) \leq \frac{1}{n^2\varepsilon^2}D(S_n) \rightarrow 0$$

于是 $\frac{1}{n}(S_n - E(S_n)) \xrightarrow{\mathbb{P}} 0$. □

Corollary 0.4.2 若 $\{\xi_n\}$ 满足 $D(\xi_n) \leq C, \forall n \geq 1$, 则 WLLN 成立.

Corollary 0.4.3 若 $\{\xi_n\}$ 为 iid, 且 $\xi_1 \sim \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix}, p \in (0, 1)$, 则 WLLN 成立.

Lemma 0.4.4 (*Borel-Cantelli Lemma, Lemma 1*)

给定概率空间 $(\Omega, \mathcal{F}, P)$, 设 $\{A_n\} \subset \mathcal{F}$

- 若 $\sum_{n=1}^{\infty} P(A_n) < +\infty$, 则 $P(\overline{\lim}_n A_n) = P(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n) = 0$;
- 若 $\{A_n\}$ 独立, 则

$$\sum_{n=1}^{\infty} P(A_n) = \infty \iff P(\overline{\lim}_n A_n) = P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n\right) = 1$$

Proof

•

$$P(\overline{\lim}_n A_n) = P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n\right) = \lim_{k \rightarrow \infty} P\left(\bigcup_{n=k}^{\infty} A_n\right) \leq \lim_{k \rightarrow \infty} \sum_{n=k}^{\infty} P(A_n) = 0$$

•

$\Leftarrow$: $\sum_{n=1}^{\infty} P(A_n)$ 收敛或发散至 ∞ , 由上式 $\lim_{k \rightarrow \infty} \sum_{n=k}^{\infty} P(A_n) \geq 1$, 故只能是发散至 ∞ .

$\Rightarrow$: 考虑 $(\overline{\lim}_n A_n)^c = \underline{\lim}_n A_n^c$. 注意对任意 k , $\sum_{n=k}^{\infty} P(A_n) = \infty$, 故

$$\begin{aligned} P\left(\underline{\lim}_n A_n^c\right) &= P\left(\bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n^c\right) = \lim_k P\left(\bigcap_{n=k}^{\infty} A_n^c\right) = \lim_k \prod_{n=k}^{\infty} P(A_n^c) \\ &= \lim_k e^{\sum_{n=k}^{\infty} \ln P(A_n^c)} = \lim_k e^{\sum_{n=k}^{\infty} \ln(1-P(A_n))} \leq \lim_k e^{-\sum_{n=k}^{\infty} P(A_n)} = 0 \end{aligned}$$

即

$$P(\overline{\lim}_n A_n) = 1$$

□

Lemma 0.4.5 (*Kolmogorov's Maximum Inq, Lemma 2*)

设 $\{\xi_k\}$ 为独立随机变量列, 且 $D(\xi_n) < +\infty$. 记 $S_n = \sum_{k=1}^n \xi_k$, 则有

$$P\left(\left\{\omega : \max_{1 \leq k \leq n} |S_k - E(S_k)| \geq \varepsilon\right\}\right) \leq \frac{D(S_n)}{\varepsilon^2} = \frac{1}{\varepsilon^2} \sum_{k=1}^n D(\xi_k)$$

进一步, 若 $|\xi_n| \leq c$, 我们还有

$$1 - \frac{(\varepsilon + 2c)^2}{D(S_n)} \leq P\left(\left\{\omega : \sup_{1 \leq k \leq n} |S_k - E(S_k)| \geq \varepsilon\right\}\right) \leq \frac{D(S_n)}{\varepsilon^2}$$

Proof

通过 $\xi'_n = \xi_n - E(\xi_n)$, 可以不妨设 $E(\xi_n) = 0$, 记 $B_k = \{\omega : |S_k| \geq \varepsilon\}$, 所证即

$$P\left(\bigcup_{k=1}^n B_k\right) \leq \frac{D(S_n)}{\varepsilon^2}$$

记 $A_1 = B_1, A_k = B_k B_{k-1}^c \cdots B_1^c$ ($k \geq 2$), 则 $\bigcup_{k=1}^n B_k = \sum_{k=1}^n A_k$. 由 A_k 的含义得

$$P(A_k) \leq \int_{A_k} \frac{S_k^2}{\varepsilon^2} dP = \frac{1}{\varepsilon^2} E(S_k^2 I_{A_k})$$

由独立性, 注意到

$$E((S_n - S_k) S_k I_{A_k}) = E(S_n - S_k) E(S_k I_{A_k}) = 0$$

于是

$$E(S_k^2 I_{A_k}) \leq E(S_k^2 I_{A_k} + (S_n - S_k)^2 I_{A_k}) = E(S_n^2 I_{A_k})$$

从而

$$P\left(\bigcup_{k=1}^n B_k\right) = \sum_{k=1}^n P(A_k) \leq \frac{1}{\varepsilon^2} \sum_{k=1}^n E(S_n^2 I_{A_k}) \leq \frac{1}{\varepsilon^2} E(S_n^2) = \frac{1}{\varepsilon^2} D(S_n)$$

另一边从没用过, 证明实在不想背了. □

Lemma 0.4.6 (*Kronecker's Lemma, Lemma 3*)

设 $\{c_n\}$ 为实数列, 记 $s_n = \sum_{k=1}^n c_k$. 则

- $c_n \rightarrow c \implies \frac{s_n}{n} \rightarrow c$;
- $\sum_n c_n$ 收敛 $\implies 0 = b := \inf_{m \geq 1} \sup_{k \geq 1} \{|s_{m+k} - s_m|\}$;
- $\sum_n \frac{c_n}{n}$ 收敛 $\implies \frac{s_n}{n} \rightarrow 0$;
- $\sum_n \frac{c_n}{a_n}$ 收敛 $\implies \frac{s_n}{a_n} \rightarrow 0$, 其中 $0 < a_n \uparrow +\infty$.

Lemma 0.4.7 (Lemma 4)

若 $\{\xi_n\}$ 与 $\{\eta_n\}$ 等价, 即 $\sum_{n=1}^{\infty} P(\{\xi_n \neq \eta_n\}) < +\infty$, 则 $\sum_{n=1}^{\infty} (\xi_n - \eta_n)$ 几乎处处收敛.

Proof

B-C 引理易证. □

Theorem 0.4.8 设 $\{\xi_n\}$ 为独立随机变量列且 $E(\xi_n^2) < \infty$, 若 $\sum_{n=1}^{\infty} D(\xi_n) < \infty$, 则 $\sum_{n=1}^{\infty} (\xi_n - E(\xi_n))$ 几乎处处收敛.

Proof

通过变换 $\eta_n = \xi_n - E(\xi_n)$, 可不妨设 $E(\xi_k) = 0$, 由 Kolmogorov 不等式知

$$P\left(\left\{\omega : \max_{M < m \leq N} |S_m - S_M| \geq \varepsilon\right\}\right) \leq \frac{1}{\varepsilon^2} \sum_{k=M+1}^N D(\xi_k)$$

$$\xrightarrow{N \rightarrow \infty} P\left(\left\{\omega : \max_{m > M} |S_m - S_M| > \varepsilon\right\}\right) \leq \frac{1}{\varepsilon^2} \sum_{k=M+1}^{\infty} D(\xi_k), \forall M \geq 1$$

记 $W_M := \max_{m, n > M} |S_m - S_n|$, 由 $|S_m - S_n| \leq |S_m - S_M| + |S_n - S_M|$ 可知

$$P(\{\omega : W_M > 2\varepsilon\}) \leq 2P\left(\left\{\omega : \max_{m > M} |S_m - S_M| > \varepsilon\right\}\right) \leq \frac{2}{\varepsilon^2} \sum_{k=M+1}^{\infty} D(\xi_k)$$

易见 $\{\omega : W_M > 2\varepsilon\}$ 随 M 递增而递减. 因此

$$P\left(\bigcap_{M=1}^{\infty} \{\omega : W_M \geq 2\varepsilon\}\right) = \lim_{M \rightarrow \infty} P(\{\omega : W_M \geq 2\varepsilon\}) = 0$$

$$\Rightarrow P\left(\bigcup_{k=1}^{\infty} \bigcap_{M=1}^{\infty} \left\{\omega : W_M \geq \frac{1}{k}\right\}\right) = 0$$

$$\Rightarrow P\left(\tilde{\Omega} := \bigcap_{k=1}^{\infty} \bigcup_{M=1}^{\infty} \left\{\omega : W_M < \frac{1}{k}\right\}\right) = 1$$

我们有 $\omega \in \tilde{\Omega} \iff \forall k \geq 1, \exists M_0 \geq 1$, s.t. $\forall n, m > M_0, |S_m - S_n| < \frac{1}{k}$. 由 Cauchy 准则即知此时 $\lim_n S_n(\omega)$ 存在, 因此 $\lim_n S_n$ 几乎处处收敛. □

Theorem 0.4.9 (Kolmogorov SLLN)

设 $\{\xi_n\}$ 为独立随机变量列, 满足 $\sum_{n=1}^{\infty} \frac{1}{n^2} D(\xi_n) < +\infty$, 则 SLLN 成立.

Proof

$$\sum_{n=1}^{\infty} D\left(\frac{\xi_n}{n}\right) = \sum_{n=1}^{\infty} \frac{1}{n^2} D(\xi_n) < +\infty \xrightarrow{\text{Thm 0.4.5}} \sum_{n=1}^{\infty} \frac{1}{n} (\xi_n - E(\xi_n)) \text{ a.s. 收敛}$$

$$\xrightarrow{\text{Lemma 3}} \frac{1}{n} (S_n - E(S_n)) \xrightarrow{\text{a.s.}} 0$$

□

Theorem 0.4.10 (iid 时的 SLLN)

设 $\{\xi_n\}$ 为 iid

- 若 $E(|\xi_1|) < +\infty$, 则 SLLN 成立, 即 $X_n := E(\xi_1|S_n) = \frac{1}{n}S_n \xrightarrow{\text{a.s.}} a = E(\xi_1)$.
- 若 $E(|\xi_1|) = +\infty$, 则

$$\overline{\lim}_n \frac{|S_n|}{n} = +\infty \text{ a.s.}$$

Proof

先看 $E(|\xi_1|) < +\infty$ 的情形. 令 $\eta_n := \xi_n \cdot I_{[-n,n]}(\xi_n)$ (对 ξ_n 截断), 则 $|\eta_n| \leq n$. 记

$$S_n = \sum_{k=1}^n \xi_k, \quad S_n^* = \sum_{k=1}^n \eta_k$$

于是

$$\frac{1}{n}S_n = \frac{1}{n}(S_n^* - E(S_n^*)) + \frac{1}{n}(S_n - S_n^*) + \frac{1}{n}E(S_n^*)$$

- 对于 $\frac{1}{n}E(S_n^*) = \frac{1}{n} \sum_{k=1}^n E(\eta_k)$, 我们有

$$\lim_{k \rightarrow \infty} E(\eta_k) = \lim_{k \rightarrow \infty} \int_{\mathbb{R}} x \cdot I_{[-k,k]}(x) dF_{\xi_1}(x) \stackrel{\text{DCT}}{=} \int_{\mathbb{R}} x dF_{\xi_1}(x) = E(\xi_1) = a < \infty$$

由 Kronecker's 引理即得 $\frac{1}{n}E(S_n^*) \rightarrow a$.

- 为证明 $\frac{1}{n}(S_n^* - E(S_n^*)) \xrightarrow{\text{a.s.}} 0$, 由 Kolmogorov SLLN, 只需证明

$$\sum_{n=1}^{\infty} \frac{D(\eta_n)}{n^2} < +\infty$$

我们有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{D(\eta_n)}{n^2} &\leq \sum_{n=1}^{\infty} \frac{E(\eta_n^2)}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} \int_{\mathbb{R}} x^2 I_{[-n,n]}(x) dF_{\xi_1}(x) \\ &= \sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{j=1}^n \int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{\xi_1}(x) \\ (\text{Fubini}) &= \sum_{j=1}^{\infty} \left(\int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{\xi_1}(x) \sum_{n=j}^{\infty} \frac{1}{n^2} \right) \\ &\leq \sum_{j=1}^{\infty} \frac{c}{j} \int_{\{j-1 \leq |x| \leq j\}} x^2 dF_{\xi_1}(x) \\ &\leq c \sum_{j=1}^{\infty} \int_{\{j-1 \leq |x| \leq j\}} |x| dF_{\xi_1}(x) \end{aligned}$$

$$= cE(|\xi_1|) < +\infty \implies \frac{1}{n}(S_n^* - E(S_n^*)) \xrightarrow{\text{a.s.}} 0$$

- 为证明 $\frac{1}{n}(S_n - S_n^*) \xrightarrow{\text{a.s.}} 0$, 我们有

$$\begin{aligned} \sum_{n=1}^{\infty} P(\{\xi_n \neq \eta_n\}) &= \sum_{n=1}^{\infty} P(\{\omega : |\xi_n| \geq n\}) \stackrel{\text{iid}}{=} \sum_{n=1}^{\infty} P(\{\omega : |\xi_1| > n\}) \\ &\text{(from KLC 3.2.4)} \leq E(|\xi_1|) < +\infty \end{aligned}$$

由 Lemma 4, 我们有

$$\sum_{n=1}^{\infty} (\xi_n - \eta_n) \text{ a.s. converges} \implies \xi_n - \eta_n \xrightarrow{\text{a.s.}} 0 \stackrel{\text{Lemma 3}}{\implies} \frac{1}{n}(S_n - S_n^*) \xrightarrow{\text{a.s.}} 0$$

综上可知 $\frac{1}{n}S_n \xrightarrow{\text{a.s.}} a$.

若 $E(|\xi_1|) = +\infty$, 则对任意 $A > 0$, 有 $E\left(\frac{|\xi_1|}{A}\right) = +\infty$, 由钟开莱的习题 3.2.4 即知

$$\sum_{n=1}^{\infty} P\{|\xi_1| \geq An\} = +\infty \stackrel{\text{iid}}{\implies} \sum_{n=1}^{\infty} P\{|\xi_n| \geq An\} = +\infty$$

从而由 B-C 引理知 $P\left\{\overline{\lim}_n \{|\xi_n| \geq An\}\right\} = 1$. 再由

$$\{|S_n - S_{n-1}| = |\xi_n| \geq An\} \subset \left\{|S_n| \geq \frac{1}{2}An\right\} \cup \left\{\left|S_{n-1}\right| \geq \frac{1}{2}A(n-1)\right\}$$

即得

$$P\left\{\overline{\lim}_n \left\{|S_n| \geq \frac{1}{2}An\right\}\right\} = 1$$

因此对任意 $A > 0$, 存在事件 Z_A 使得 $P\{Z_A\} = 0$ 且对于 $\omega \in Z_A^c$ 有

$$\overline{\lim}_n \frac{S_n(\omega)}{n} \geq \frac{A}{2}$$

令 $Z = \bigcup_{m=1}^{\infty} Z_m$, 于是 $P\{Z\} = 0$, 且对任意 $\omega \in Z^c$, 有

$$\overline{\lim}_n \frac{S_n(\omega)}{n} \geq \frac{m}{2}, \forall m \geq 1$$

令 $m \rightarrow \infty$ 即知 $\overline{\lim}_n \frac{S_n(\omega)}{n} = \infty$ a.s. □

Remark

这位更是重量级.

0.5 中心极限定理

符号设置: 对期望方差均有限的独立随机变量列 $\{\xi_n\}$, 记 ℓ_n 为其标准化, 并记

$$\ell_n := \frac{S_n - E(S_n)}{\sqrt{D(S_n)}} = \sum_{k=1}^n \frac{\xi_k - E(\xi_k)}{\sqrt{D(S_n)}} := \sum_{k=1}^n \xi_{nk} \text{ with } \xi_{nk} := \frac{\xi_k - E(\xi_k)}{\sqrt{D(S_n)}}, 1 \leq k \leq n$$

$$\sigma_k^2 = D(\xi_k), \quad a_k = E(\xi_k), \quad B_n^2 = D(S_n) = \sum_{k=1}^n \sigma_k^2$$

$$E(\xi_{nk}) = 0, \quad D(\xi_{nk}) = \frac{\sigma_k^2}{B_n^2} =: \sigma_{nk}^2 \implies \sum_{k=1}^n \sigma_{nk}^2 = 1$$

Theorem 0.5.1 (1) (Def of CLT) $\ell_n \xrightarrow{d} \xi \sim N(0, 1)$, i.e. $f_{\ell_n}(t) \xrightarrow{w} f_\xi(t) = e^{-\frac{1}{2}t^2}, \forall t$;
(2) (Feller's condition)

$$\lim_n \max_{1 \leq j \leq n} \sigma_{nj}^2 = \lim_n \max_{1 \leq j \leq n} \frac{\sigma_j^2}{B_n^2} = 0$$

(3) (Linderberg condition)

$$0 = \lim_n \frac{1}{B_n^2} \sum_{k=1}^n \int_{\frac{|x-a_k|}{B_n} \geq \tau} (x - a_k)^2 dF_{\xi_k}(x) = \lim_n \sum_{k=1}^n \int_{|x| \geq \tau} x^2 dF_{\xi_{nk}}(x), \forall \tau > 0$$

那么 (1)+(2) $\iff$ (3).

Remark

$$(2') \lim_n \max_{1 \leq j \leq n} P(\{\omega : |\xi_{nj}| \geq \varepsilon\}) = 0, \forall \varepsilon > 0$$

$$(2'') \lim_n P\left(\left\{\omega : \max_{1 \leq j \leq n} |\xi_{nj}| \geq \varepsilon\right\}\right) = 0, \forall \varepsilon > 0$$

$$(1)+(2') \iff (3) \iff (1)+(2'')$$

Proof

只证充分性. 先从 Linderberg 条件推出 Feller 条件. 由 Chebyshev Inq 知

$$\begin{aligned} P(\{|\xi_{nj}| \geq \varepsilon\}) &\leq \frac{\sigma_{nj}^2}{\varepsilon^2} = \frac{1}{\varepsilon^2} \left(\int_{|x| < \tau} x^2 dF_{\xi_{nj}} + \int_{|x| \geq \tau} x^2 dF_{\xi_{nj}} \right) \\ &\leq \frac{1}{\varepsilon^2} \left(\tau^2 + \int_{|x| \geq \tau} x^2 dF_{\xi_{nj}} \right) \end{aligned}$$

这意味着

$$\max_{i \leq j \leq n} \sigma_{nj}^2 \leq \tau^2 + \sum_{j=1}^n \int_{|x| \geq \tau} x^2 dF_{\xi_{nj}} \quad \text{for (2)}$$

$$\max_{1 \leq j \leq n} P(\{|\xi_{nj}| > \varepsilon\}) \leq \frac{\tau^2}{\varepsilon^2} + \frac{1}{\varepsilon^2} \sum_{j=1}^n \int_{|x| \geq \tau} x^2 dF_{\xi_{nj}} \quad \text{for (2')}$$

$$P\left(\left\{\max_{1 \leq k \leq n} |\xi_{nk}| \geq \varepsilon\right\}\right) \leq \sum_{j=1}^n P(\{|\xi_{nj}| \geq \varepsilon\}) \leq \frac{1}{\varepsilon} \sum_{j=1}^n \int_{|x| \geq \varepsilon} x^2 dF_{\xi_{nj}} \quad \text{for (2'')}$$

先令 $n \rightarrow \infty$, 再令 $\tau \rightarrow 0$, 由 (3) 即得 (2), (2'), (2'').

再从 Linderberg 条件推出 CLT, 由数学分析小工具, 我们有

$$\begin{aligned} |f_{\ell_n}(t) - e^{-\frac{1}{2}t^2}| &= \left| \prod_{k=1}^n f_{\xi_{nk}}(t) - \prod_{k=1}^n e^{-\frac{1}{2} \frac{\sigma_k^2}{B_n^2} t^2} \right| \\ &\leq \left| \prod_{k=1}^n f_{\xi_{nk}}(t) - \prod_{k=1}^n \left(1 - \frac{1}{2} \frac{\sigma_k^2}{B_n^2} t^2\right) \right| + \left| \prod_{k=1}^n \left(1 - \frac{1}{2} \frac{\sigma_k^2}{B_n^2} t^2\right) - \prod_{k=1}^n e^{-\frac{1}{2} \frac{\sigma_k^2}{B_n^2} t^2} \right| \\ &= \square_1(n) + \square_2(n) \end{aligned}$$

依次估计 $\square_1(n)$ 与 $\square_2(n)$, 我们有

$$\begin{aligned} \square_1(n) &\leq \sum_{k=1}^n \left| f_{\xi_{nk}}(t) - \left(1 - \frac{1}{2} t^2 \frac{\sigma_k^2}{B_n^2}\right) \right| \\ &= \sum_{k=1}^n \left| E\left(e^{it\xi_{nk}} - 1 - it\xi_{nk} + \frac{1}{2} t^2 \xi_{nk}^2\right) \right| \\ &\leq \sum_{k=1}^n E(\min\{|t\xi_{nk}|^2, |t\xi_{nk}|^3\}) \\ &\leq \sum_{k=1}^n \int_{|x| \geq \tau} |tx|^2 dF_{\xi_{nk}} + \sum_{k=1}^n \int_{|x| \leq \tau} |tx|^3 dF_{\xi_{nk}}, \quad \forall \tau > 0 \\ &\leq t^2 \sum_{k=1}^n \int_{|x| \geq \tau} x^2 dF_{\xi_{nk}} + |t|^3 \tau \sum_{k=1}^n \int_{\mathbb{R}} x^2 dF_{\xi_{nk}} \\ &= t^2 \sum_{k=1}^n \int_{|x| \geq \tau} x^2 dF_{\xi_{nk}} + |t|^3 \tau \end{aligned}$$

先令 $n \rightarrow \infty$, 再令 $\tau \rightarrow 0$, 我们得到 $\square_1(n) \rightarrow 0$.

$$\begin{aligned} \square_2(n) &\leq \sum_{k=1}^n \left| e^{-\frac{1}{2} t^2 \frac{\sigma_k^2}{B_n^2}} - \left(1 - \frac{1}{2} t^2 \frac{\sigma_k^2}{B_n^2}\right) \right| \\ &\leq \sum_{k=1}^n \left| -\frac{1}{2} t^2 \frac{\sigma_k^2}{B_n^2} \right|^2 \leq \frac{1}{4} t^4 \max_{1 \leq k \leq n} \sigma_{nk}^2 \xrightarrow{(2)} 0 \quad (t \text{ 固定, 令 } n \rightarrow \infty) \end{aligned}$$

由如上两个估计, 我们最终得到 $f_{\ell_n}(t) \xrightarrow{w} e^{-\frac{1}{2}t^2}$. □

重要习题:

Problem

设 $\{\xi_n\}$ 为独立随机变量序列, 每个 ξ_n 有分布 $P\{\xi_n = \pm n^\alpha\} = \frac{1}{2}$. 试证:

(1) 当 $\alpha > -\frac{1}{2}$ 时, $\{\xi_n\}$ 满足 CLT;

(2) 当 $\alpha > 0$ 时, $\{\xi_n\}$ 服从大数定律的充要条件是 $\alpha < \frac{1}{2}$.

Proof

(1) 易见 $E(\xi_k) = 0, D(\xi_k) = n^{2\alpha}, D(S_n) = \sum_{k=1}^n k^{2\alpha}$, 从而由 $2\alpha > -1$, 选取 $\delta > 0$ 使得 $\alpha(2+\delta) \neq -1$, 于是

$$\frac{1}{B_n^{2+\delta}} \sum_{k=1}^n E(|\xi_k|^{2+\delta}) = \frac{\sum_{k=1}^n k^{\alpha(2+\delta)}}{\left(\sum_{k=1}^n k^{2\alpha}\right)^{1+\frac{\delta}{2}}} \approx \frac{n^{2\alpha+\alpha\delta+1}}{n^{2\alpha+\alpha\delta+1+\frac{\delta}{2}}} \rightarrow 0 (n \rightarrow \infty)$$

从而 $\{\xi_n\}$ 满足 Lyapunov 条件, 故满足 CLT.

(2) 由(1), 此时 $\{\xi_n\}$ 为独立的满足 CLT 的随机变量列, 故由 4.3.6.8 可知 $\{\xi_n\}$ 满足大数定律当且仅当 $\frac{1}{n^2} D(S_n) = \frac{1}{n^2} \sum_{k=1}^n k^{2\alpha} \approx n^{2\alpha-1} \rightarrow 0$, 显然这等价于 $\alpha < \frac{1}{2}$.

Problem

设 $\{\xi_n\}$ 为 iid, $E(\xi_n) = 0, D(\xi_n) = 1$. 试证如下随机变量列为渐近正态分布的

$$(1) \eta_n = \sqrt{n} \frac{\xi_1 + \dots + \xi_n}{\xi_1^2 + \dots + \xi_n^2} \quad (2) \zeta_n = \frac{\xi_1 + \dots + \xi_n}{\sqrt{\xi_1^2 + \dots + \xi_n^2}}$$

Proof

(1) 记 $S_n = \sum_{k=1}^n \xi_k, T_n = \sum_{k=1}^n \xi_k^2$, 则 $E(S_n) = 0, D(S_n) = E(T_n) = n$. 再由 $\{\xi_n\}, \{\xi_n^2\}$ 均为 iid, 且 ξ_1 期望方差有限, ξ_1^2 期望有限, 故 $\{\xi_n\}$ 满足 CLT, $\{\xi_n^2\}$ 满足 SLLN, 特别地, 其满足 WLLN, 即

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1), \quad \frac{T_n}{n} \xrightarrow{\mathbb{P}} 1 \implies \frac{n}{T_n} \xrightarrow{\mathbb{P}} 1 \implies \eta_n = \frac{S_n}{\sqrt{n}} \frac{n}{T_n} \xrightarrow{d} N(0, 1)$$

(2) 同理

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, 1), \quad \sqrt{\frac{n}{T_n}} \xrightarrow{\mathbb{P}} 1 \implies \zeta_n = \frac{S_n}{\sqrt{n}} \sqrt{\frac{n}{T_n}} \xrightarrow{d} N(0, 1)$$

□

Problem

设独立随机变量列 $\{\xi_n\}$ 有分布

$$P\{\xi_n = \pm n\} = \frac{1}{2n^2}, \quad P\{\xi_n = 0\} = 1 - \frac{1}{n^2}$$

求证: $\{\xi_n\}$ 不满足 CLT.

Proof

易见 $E(\xi_k) = 0, D(\xi_k) = 1, D(S_n) = n \Rightarrow \lim_n \max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = 0$. 从而 $\{\xi_n\}$ 满足 Feller 条件, 若其满足 CLT, 则必然满足 Linderberg 条件. 我们计算

$$\begin{aligned} \frac{1}{B_n^2} \sum_{k=1}^n \int_{\frac{|x-a_k|}{B_n} > \tau} (x-a_k)^2 dF_{\xi_k}(x) &= \frac{1}{B_n^2} \sum_{k=1}^n E(\xi_k^2 I_{|\xi_k| > \tau B_n}) \\ &= \frac{1}{n} \sum_{k=\lceil \tau n \rceil}^n 1 = 1 + \frac{1}{n} - \frac{[\tau n]}{n} \\ &\approx 1 - \tau + \frac{1}{n} \end{aligned}$$

于是令 $n \rightarrow \infty$, 上式趋于 $1 - \tau \neq 0$ ($\tau \neq 1$ 时), 矛盾! 从而 $\{\xi_n\}$ 不满足 CLT. $\square$

Remark

这是一个满足 Feller 条件, 但不满足 CLT 和 Linderberg 条件的例子.

Problem

设 $\{\xi_n\}$ 为独立随机变量列, $\xi_n \sim N(0, \sigma_n^2)$, 其中 $\sigma_1^2 = 1, \sigma_n^2 = nB_{n-1}^2$. 试证 $\{\xi_n\}$ 满足 CLT 但不满足 Feller 条件.

Proof

我们有 $B_n^2 = \sigma_n^2 + B_{n-1}^2 = (n+1)B_{n-1}^2 = \cdots = \frac{1}{2}(n+1)!, \sigma_n^2 = \frac{n}{2}n!$, 从而

$$\max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = \max_{1 \leq k \leq n} \frac{k \cdot k!}{(n+1)!} = \frac{n}{n+1} \Rightarrow \lim_n \max_{1 \leq k \leq n} \frac{\sigma_k^2}{B_n^2} = 1$$

故 $\{\xi_n\}$ 不满足 Feller 条件. 我们有

$$S_n \sim N(0, B_n^2) \Rightarrow \frac{S_n - E(S_n)}{\sqrt{D(S_n)}} \sim N(0, 1)$$

从而 $\{\xi_n\}$ 满足 CLT. $\square$

Remark

这是一个满足 CLT, 但不满足 Feller 条件和 Linderberg 条件的例子.

Problem

设 $\{\xi_{nk}, 1 \leq k \leq n\}$ 独立, 满足 $\xi_{nk} \sim \begin{pmatrix} 0 & 1 \\ 1-p_{nk} & p_{nk} \end{pmatrix}$. 记 $S_n = \sum_{k=1}^n \xi_k$. 证明

(1) 若 $\frac{1}{n} \sum_{k=1}^n p_{nk} \rightarrow p$ ($n \rightarrow \infty$), 则 $\frac{1}{n} S_n \xrightarrow{\text{a.s.}} p$;

(2) 若 $\max_{1 \leq k \leq n} p_{nk} \rightarrow 0$ ($n \rightarrow \infty$) 且 $\sum_{k=1}^n p_{nk} \rightarrow \lambda > 0$, 则 $S_n \xrightarrow{d} S \sim \text{Poisson}(\lambda)$.

Proof

(1) 易见 $E(\xi_{nk}) = p_{nk}$, $D(\xi_{nk}) = p_{nk} - p_{nk}^2$. 对 $\varepsilon > 0$, 由 Chebyshev's Inq

$$\begin{aligned} P\left\{\left|\frac{S_n - E(S_n)}{n}\right| \geq \varepsilon\right\} &\leq \frac{E(S_n - E(S_n))^4}{n^4 \varepsilon^4} = \frac{1}{n^4 \varepsilon^4} E\left[\sum_{k=1}^n (\xi_{nk} - E(\xi_{nk}))\right]^4 \\ &= \frac{1}{n^4 \varepsilon^4} \left(\sum_{k=1}^n E(\xi_{nk} - E(\xi_{nk}))^4 + 6 \sum_{1 \leq i < j \leq n} E[(\xi_{ni} - E(\xi_{ni}))^2 (\xi_{nj} - E(\xi_{nj}))^2] \right) \end{aligned}$$

上述最后一个展开需要使用 $\{\xi_{nk}\}$ 的独立性. 由于 $|\xi_{nk} - E(\xi_{nk})| \in [0, 1]$, 故有

$$P\left\{\left|\frac{S_n - E(S_n)}{n}\right| > \varepsilon\right\} \leq \frac{1}{n^4 \varepsilon^4} \left(n + 6 \cdot \frac{n(n-1)}{2} \right) \approx \frac{1}{n^2}$$

从而由 $\sum \frac{1}{n^2} < \infty$ 以及 Borel-Cantelli 引理即知

$$P\left\{\overline{\lim}_n \left\{\left|\frac{S_n - E(S_n)}{n}\right| > \varepsilon\right\}\right\} = 0 \Rightarrow \frac{S_n - E(S_n)}{n} \xrightarrow{\text{a.s.}} 0$$

于是 $\frac{E(S_n)}{n} = \frac{1}{n} \sum_{k=1}^n p_{nk} \rightarrow p \Rightarrow \frac{S_n}{n} \xrightarrow{\text{a.s.}} p$.

(2) 由独立性可知

$$f_{S_n}(t) = \prod_{k=1}^n E(e^{it\xi_{nk}}) = \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}), \quad f_S(t) = e^{\lambda(e^{it}-1)}$$

根据证明 CLT 时用到的引理, 我们有

$$\begin{aligned} |f_{S_n}(t) - f_S(t)| &= \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - e^{\lambda(e^{it}-1)} \right| \\ &= \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - e^{\sum_{k=1}^n p_{nk}(e^{it}-1)} + e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \left| \prod_{k=1}^n (1 - p_{nk} + p_{nk}e^{it}) - \prod_{k=1}^n e^{p_{nk}(e^{it}-1)} \right| + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \sum_{k=1}^n |e^{p_{nk}(e^{it}-1)} - 1 - p_{nk}(e^{it}-1)| + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq \sum_{k=1}^n p_{nk}^2 |e^{it} - 1|^2 + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \\ &\leq 4 \max_{1 \leq k \leq n} p_{nk} \sum_{k=1}^n p_{nk} + \left| e^{(e^{it}-1)\sum_{k=1}^n p_{nk}} - e^{\lambda(e^{it}-1)} \right| \rightarrow 0 \quad (n \rightarrow \infty) \end{aligned}$$

由 t 的任意性, 令 $n \rightarrow \infty$ 即知命题成立

□

Problem

(非标准 CLT)

设 $\{\xi_n\}$ 为 iid 且 $E(\xi_1) = 0, D(\xi_1) = 1$. v_n 为整值随机变量, $\frac{v_n}{n} \xrightarrow{\mathbb{P}} c \in (0, \infty)$, 则

$$\frac{S_{v_n}}{\sqrt{v_n}} \xrightarrow{d} \xi \sim N(0, 1) \text{ where } S_n = \sum_{k=1}^n \xi_k$$

Proof

利用分解

$$\frac{S_{v_n}}{\sqrt{v_n}} = \left(\frac{S_{[nc]}}{\sqrt{[nc]}} + \frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \right) \sqrt{\frac{[nc]}{v_n}}$$

由经典的 CLT 以及依概率收敛的相关性质, 我们有

$$\frac{S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{d} \xi, \quad \frac{[nc]}{v_n} = \frac{nc}{v_n} \frac{[nc]}{nc} \xrightarrow{\mathbb{P}} 1 \left(\Leftrightarrow \xrightarrow{d} 1 \right) \Rightarrow \sqrt{\frac{[nc]}{v_n}} \xrightarrow{\mathbb{P}} 1 \left(\Leftrightarrow \xrightarrow{d} 1 \right)$$

如果有

$$\frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{\mathbb{P}} 0 \left(\Leftrightarrow \xrightarrow{d} 0 \right) \sim (1)$$

于是由中文教材作业题 4.1.4.11 和 4.1.4.12 知

$$\frac{S_{[nc]}}{\sqrt{[nc]}} + \frac{S_{v_n} - S_{[nc]}}{\sqrt{[nc]}} \xrightarrow{d} \xi \Rightarrow \frac{S_{v_n}}{\sqrt{v_n}} \xrightarrow{d} \xi$$

下证 (1) 成立. 固定 $0 < \varepsilon < 1$, 由于 $\frac{v_n}{[cn]} \xrightarrow{\mathbb{P}} 1$, 故存在 N 使得对 $n > N$ 有

$$P \left\{ \left| \frac{v_n}{[cn]} - 1 \right| > \varepsilon^3 \right\} < \varepsilon$$

记 $a_n = \lfloor (1 - \varepsilon^3)[cn] \rfloor, b_n = \lfloor (1 + \varepsilon^3)[cn] \rfloor$, 记 $\Lambda := \{\omega : a_n \leq v_n(\omega) \leq b_n\}$, 若 $\omega \in \Lambda$, 则 $S_{v_n(\omega)}(\omega)$ 是某个和式 S_j , 其中 $a_n \leq j \leq b_n$. 对 $[cn] < j \leq b_n$, 有

$$S_j - S_{[cn]} = X_{[cn]+1} + X_{[cn]+2} + \cdots + X_j$$

于是由 Kolmogorov 不等式可知

$$P \left\{ \max_{[cn] \leq j \leq b_n} |S_j - S_{[cn]}| > \varepsilon [cn] \right\} \leq \frac{D(S_{b_n} - S_{[cn]})}{\varepsilon^2 cn} = \frac{b_n - [cn]}{\varepsilon^2 cn} \leq \frac{\varepsilon^3 [cn]}{\varepsilon^2 cn} \leq \varepsilon$$

对 $a_n \leq j < [cn]$, 同理可知

$$P\left\{\max_{a_n \leq j < \lfloor cn \rfloor} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \lfloor cn \rfloor\right\} \leq \frac{D(S_{\lfloor cn \rfloor} - S_{a_n})}{\varepsilon^2 cn} \leq \varepsilon$$

结合以上两个不等式可知

$$P\left\{\max_{a_n \leq j \leq b_n} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \lfloor cn \rfloor\right\} \leq 2\varepsilon$$

故对 $n > N$, 我们有

$$\begin{aligned} P\left\{\left|\frac{S_{v_n} - \lfloor cn \rfloor}{\sqrt{\lfloor cn \rfloor}}\right| > \varepsilon\right\} &= \sum_{j=1}^{\infty} P\left\{v_n = j; \left|\frac{S_{v_n} - S_{\lfloor cn \rfloor}}{\sqrt{\lfloor cn \rfloor}}\right| > \varepsilon\right\} \\ &\leq \sum_{a_n \leq j \leq b_n} P\left\{v_n = j; \max_{a_n \leq j \leq b_n} |(S_j - S_{\lfloor cn \rfloor})| > \varepsilon \sqrt{\lfloor cn \rfloor}\right\} + \sum_{j \notin [a_n, b_n]} P\{v_n = j\} \\ &\leq P\left\{\max_{a_n \leq j \leq b_n} |S_j - S_{\lfloor cn \rfloor}| > \varepsilon \sqrt{\lfloor cn \rfloor}\right\} + P\{v_n \notin [a_n, b_n]\} \leq 2\varepsilon + \varepsilon = 3\varepsilon \end{aligned}$$

由 ε 的任意性可知 (1) 成立, 从而原命题成立. □

Corollary 0.5.2 (Lyapunov's Thm)

设 $\{\xi_n\}$ 独立并且存在 $\delta > 0$ 使得

$$\lim_n \frac{1}{B_n^{2+\delta}} \sum_{k=1}^n E(|\xi_k - E(\xi_k)|^{2+\delta}) = 0$$

则 CLT 成立.

Corollary 0.5.3 设 $\{\xi_n\}$ 独立且满足 $\max_{1 \leq k \leq n} |\xi_k| \leq L_n$, $\lim_n \frac{L_n}{B_n} = 0$, 则 CLT 成立.

0.6 鞅

Example

$B = \{B_t, t \geq 0\}$ is Brownian Motion (BM) on $(\Omega, \mathcal{F}, P)$, $\mathcal{F}_t = \sigma(B_s, s \leq t)$. Then

- B is a martingale on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$;
- $\{e^{iB_t + \frac{1}{2}t}, t \geq 0\}$ is a martingale on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$;
- $\{B_t^2 - t\}$ is a martingale on $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, P)$.

Proof

期望有限和适应性都是显然的, 只证三个鞅性.

(1) 对任意 $t > s$, 我们有

$$E(B_t | \mathcal{F}_s) - B_s = E((B_t - B_s) | \mathcal{F}_s) \stackrel{\text{indep}}{=} E(B_t - B_s) = E(N(0, t-s)) = 0$$

(2) 对任意 $t > s$, 我们有

$$\begin{aligned} E\left(e^{iB_t + \frac{1}{2}t} | \mathcal{F}_s\right) - \left(e^{iB_s + \frac{1}{2}s}\right) &= E\left(\left(e^{iB_t + \frac{1}{2}t} - e^{iB_s + \frac{1}{2}s}\right) | \mathcal{F}_s\right) \\ &= e^{iB_s + \frac{1}{2}s} E\left(\left(e^{i(B_t - B_s) + \frac{1}{2}(t-s)} - 1\right) | \mathcal{F}_s\right) \\ &\stackrel{\text{indep}}{=} e^{iB_s + \frac{1}{2}s} E\left(e^{i(B_t - B_s) + \frac{1}{2}(t-s)} - 1\right) \\ &= e^{iB_s + \frac{1}{2}s} \left(e^{\frac{1}{2}(t-s)} f_{N(0, t-s)}(1) - 1\right) = 0 \end{aligned}$$

(3) 对任意 $t > s$, 我们有

$$\begin{aligned} E(B_t^2 - t | \mathcal{F}_s) - (B_s^2 - s) &= E(B_t^2 - B_s^2 | \mathcal{F}_s) - (t - s) \\ E(B_t B_s | \mathcal{F}_s) &= B_s E(B_t | \mathcal{F}_s) \stackrel{(1)}{=} B_s^2 = E(B_s^2 | \mathcal{F}_s) \\ \Rightarrow E(B_t^2 - B_s^2 | \mathcal{F}_s) &= E((B_t - B_s)^2 | \mathcal{F}_s) \stackrel{\text{indep}}{=} E((B_t - B_s)^2) = t - s \quad \square \end{aligned}$$

Remark

Review: **Brownian Motion** (BM) on $(\Omega, \mathcal{F}, P)$: $B = \{B_t, t \geq 0\}$ satisfies:

- i) B_t 关于 t 几乎处处连续;
- ii) $B_t - B_s \sim N(0, t-s)$, $\forall 0 \leq s \leq t$;
- iii) $B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent for any $0 \leq t_0 < t_1 < \dots < t_n < +\infty$
- iv) $B_0 \stackrel{a.s.}{=} 0$

Remark

- ξ 在 $\mathcal{F}$ 上可测, 则 $E(\xi\eta | \mathcal{F}) = \xi E(\eta | \mathcal{F})$;
- ξ 与 $\mathcal{F}$ 独立, 则 $E(\xi | \mathcal{F}) = E(\xi)$.

1 后记-2025 春季学期 lzx 概率论期末考试

每题 20 分, 取得分最多的 5 道题计算总分. 约定 $(\Omega, \mathcal{F}, P)$ 为概率空间, $\{\xi_n\}$ 为随机变量.

1.1 Problem 1

- (1) 证明: $\xi_n \xrightarrow{\mathbb{P}} \xi \implies \xi_n \xrightarrow{d} \xi$. 反之是否成立? 若成立, 给出证明; 若不成立, 给出反例.
- (2) 证明: $\xi_n \xrightarrow{\mathbb{P}} c \iff \xi_n \xrightarrow{d} c$, 其中 c 为常数.
- (3) 设 $\eta_n \xrightarrow{d} 0, \xi_n \xrightarrow{d} \xi$, 证明: $\eta_n \xi_n \xrightarrow{d} 0$.
- (4) 证明: $\sum_{n=1}^{\infty} P(A_n) < \infty \implies P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = 0$.

1.2 Problem 2

设 $\{\xi_n\}$ 为 iid, $\xi_1 \sim \text{Exp}(1)$, $N \sim G(p)$, 且 $\{\xi_n\}$ 与 N 独立, 记 $X = \sum_{k=1}^N \xi_k$, 求:

- (1) X 的分布函数; (2) $E(X)$; (3) $D(X)$.

1.3 Problem 3

设 $\{\xi_n\}$ 为 iid, $\xi_1 \sim \text{Exp}(1)$, 证明:

- (1) $P\left\{\omega : \lim_n \frac{M_n}{\log n} = 1\right\} = 1$, 其中 $M_n = \max\{\xi_1, \dots, \xi_n\}$.
- (2) 若 $\alpha > 1$, 则 $P\left\{\overline{\lim}_n \{\omega : \xi_n > \alpha \log n\}\right\} = 0$.
- (3) 若 $\alpha \leq 1$, 则 $P\left\{\overline{\lim}_n \{\omega : \xi_n > \alpha \log n\}\right\} = 1$.

1.4 Problem 4

设 $\{F_n\}, F$ 为分布函数, 记特征函数 $f_n(t) = \int_{\mathbb{R}} e^{itx} dF_n(x)$, $f(t) = \int_{\mathbb{R}} e^{itx} dF(x)$. 证明:
 $\forall t, f_n(t) \rightarrow f(t) \implies F_n \xrightarrow{w} F$.

1.5 Problem 5

设 $\{\xi_n\}$ 独立, 记 $S_n = \sum_{k=1}^n \xi_k$, 满足 $0 < D(\xi_n) < \infty, |\xi_n| \leq c$, 证明:

$$1 - \frac{(\varepsilon + 2c)^2}{D(S_n)} \leq P\left\{\max_{1 \leq j \leq n} |S_j - E(S_j)| > \varepsilon\right\} \leq \frac{D(S_n)}{\varepsilon^2}, \quad \forall \varepsilon > 0$$

1.6 Problem 6

设 $\{\xi_n\}$ 独立, $E(\xi_n) = 0$.

- (1) 证明: 若 $\sum_{k=1}^{\infty} D(\xi_k) < \infty$, 则 $\sum_{k=1}^{\infty} \xi_k$ a.s. 收敛.
- (2) 证明: 若 $\sum_{k=1}^{\infty} \frac{D(\xi_k)}{k^2} < \infty$, 则 $\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0$, 其中 $S_n = \sum_{k=1}^n \xi_k$.
- (3) 记 $\eta_n := \xi_n I_{[-A, A]}(\xi_n)$, 其中 A 为固定的非负实数, 证明: $\sum_{n=1}^{\infty} \xi_n$ a.s. 收敛可推出

$$\sum_{n=1}^{\infty} P\{\xi_n \neq \eta_n\} < \infty, \quad \sum_{n=1}^{\infty} E(\eta_n) < \infty, \quad \sum_{n=1}^{\infty} D(\eta_n) < \infty$$

1.7 Problem 7

设对每个 $n \geq 1$, $\{\xi_{n,m} : m = 1, 2, \dots, n\}$ 是独立随机变量, 满足:

(1) $E(\xi_{n,m}) = 0$;

(2) $\lim_{n \rightarrow \infty} \sum_{m=1}^n E(\xi_{n,m}) = 1$;

(3) 对任意 $\varepsilon > 0$, $\lim_{n \rightarrow \infty} \sum_{m=1}^n E(\xi_{n,m}^2 I_{\{|\xi_{n,m}| > \varepsilon\}}) = 0$.

令 $S_n := \sum_{m=1}^n \xi_{n,m}$, 证明: $S_n \xrightarrow{d} \xi \sim N(0, 1)$.

1.8 Problem 8

设 $\{Z_n\}$ 为 $(\Omega, \mathcal{F}, P)$ 上实值可积的独立随机变量, $E(Z_n) = 1$. 记 $\xi_0 \equiv 1$, $\xi_n = \prod_{k=1}^n Z_k$, $n \geq 1$, 再记 $\mathcal{F}_n = \sigma(\xi_k, 0 \leq k \leq n)$, $n \geq 0$, 证明: $\{\xi_n\}$ 是 $\{\Omega, \mathcal{F}, \{\mathcal{F}_n\}_{n \geq 0}, P\}$ 上的鞅.

一些碎碎念()

第一题是经典的随机变量四种收敛和 B-C 引理; 第二题是简单的计算题, 用全概率公式和 Taylor 公式就能很快解决; 第三题考场上没做, 疑似是个 lzx 自己手写的作业题; 第四题是连续性定理很长的那半边的证明; 第五题是双边的 Kolmogorov 不等式, 没人能想到他会考左半边; 第六题前两问是经典的 SLLN 结论, 第三问是三级数定理; 第七题也是经典的 CLT 了, 需要注意的是条件 (2) 跟课堂讲得略有不同, 需要在估计的时候再多拆出来一项; 第八题是简单的鞅, 这学期因为放假少了堂课, 所以鞅这块只讲了定义, 自然也就只能考定义.

总体来说如果背诵的广度和熟练度都到位了的话, 这卷子 90+ 很轻松, 比如我就选择了 1,2,4,7,8 这五道题. 需要注意的是, 最后卷子批得很严, 因此要注意过程的书写.